Layer24: An Emotion-Aware Intelligent Productivity Balancing System Using Machine Learning

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1.ABSTRACT

Layer24 is an emotion-aware intelligent productivity balancing system designed to help users manage daily tasks efficiently while maintaining mental and physical well-being. Traditional task management systems mainly focus on scheduling and reminders, but they often ignore human emotional conditions such as stress, mood, energy level, and sleep quality. The proposed system uses machine learning techniques to analyze both user state and task-related factors to provide intelligent task balancing suggestions such as KEEP, REDUCE, or DELAY.

The system considers multiple parameters including mood, stress level, energy level, sleep hours, task priority, task difficulty, task type, and total workload. A custom dataset was created and preprocessed using data cleaning, normalization, and encoding techniques. Machine learning models were trained and evaluated to generate productivity balancing recommendations. The developed prototype provides an interactive interface where users can enter multiple tasks and receive emotionally adaptive productivity suggestions.

The proposed approach aims to create a more human-centered productivity management system that balances performance, health, and daily life activities. Experimental results demonstrate that the system can effectively identify overload conditions and provide intelligent task management recommendations.

Keywords:

Machine Learning, Productivity Balancing, Emotion-Aware System, Artificial Intelligence, Human-Centered AI

2.INTRODUCTION

In today’s fast-paced world, maintaining productivity while balancing mental and physical well-being has become a major challenge. Most existing productivity and task management applications mainly focus on scheduling tasks, setting reminders, and improving efficiency. However, these systems often ignore important human factors such as mood, stress, energy level, and sleep quality, which directly influence a person’s ability to complete tasks effectively. As a result, users may experience burnout, stress overload, poor work-life balance, and decreased productivity.

Artificial Intelligence and Machine Learning have created new opportunities for building intelligent systems that can understand user behavior and provide adaptive recommendations. By integrating emotional and contextual factors into productivity management, it becomes possible to create systems that not only improve efficiency but also support user well-being. Emotion-aware productivity systems can help users prioritize important activities, reduce overload conditions, and maintain a healthier daily routine.

This research introduces “Layer24,” an emotion-aware intelligent productivity balancing system that uses machine learning techniques to provide personalized task balancing suggestions. The proposed system considers various user state parameters such as mood, stress level, energy level, and sleep hours along with task-related factors including task type, priority, difficulty, deadline, and workload. Based on these inputs, the system predicts whether tasks should be kept, reduced, or delayed in order to maintain a balanced and productive schedule.

A custom dataset was created for training and testing the machine learning model. Data preprocessing techniques such as cleaning, normalization, and encoding were applied before model development. The system also includes an interactive prototype interface that allows users to enter multiple tasks and receive intelligent balancing recommendations. The main objective of this work is to develop a human-centered AI productivity system that promotes both productivity and well-being rather than focusing only on task completion.

3. METHODOLOGY

3.1 Data Collection

The dataset used in this research was manually created to represent realistic daily productivity and lifestyle scenarios. The dataset contains 1000 records with user state information and task-related attributes such as mood, stress level, energy level, sleep hours, task type, task difficulty, priority, deadline, and workload. The dataset was designed to support intelligent productivity balancing using machine learning techniques.

3.2 Data Preprocessing

Data preprocessing was performed to improve data quality and prepare the dataset for machine learning model training. Different preprocessing techniques such as data cleaning, encoding, and normalization were applied to ensure consistency and better model performance.

3.2.1 Data Cleaning

Data cleaning was performed to remove duplicate values, inconsistent entries, and formatting issues from the dataset. Missing values and incorrect data types were checked and corrected. This process helped improve the reliability and quality of the dataset.

3.2.2 Data Encoding

Categorical values such as mood, task layer, task type, and task difficulty were converted into numerical values using label encoding techniques. Encoding was necessary because machine learning models require numerical input for training and prediction.

3.2.3 Data Normalization

Data normalization was applied to maintain consistency among numerical features such as stress level, energy level, sleep hours, and workload hours. This process helped improve model stability and prediction performance.

3.3 Data Visualization

Data visualization techniques were used to understand patterns and relationships within the dataset. Heatmaps, graphs, and confusion matrices were generated to analyze feature relationships and evaluate model performance visually.

3.4 Model Building

Machine learning techniques were used to build the intelligent productivity balancing system. The dataset was divided into training and testing sets for model development and evaluation. The trained model predicts whether a task should be kept, reduced, or delayed based on user state and task information.

3.4.1 Machine Learning Model

A Decision Tree Classifier was used as the primary machine learning model for this research. The model was trained using encoded dataset features and tested on unseen data to evaluate prediction accuracy and intelligent task balancing performance.

3.5 Evaluation Metrics

The performance of the machine learning model was evaluated using metrics such as accuracy score, confusion matrix, and classification report. These evaluation methods helped measure prediction performance and model effectiveness.

4.RESULTS AND DISCUSSION

The proposed Layer24 system was successfully developed and tested using machine learning techniques. The trained model achieved high prediction accuracy in identifying whether tasks should be kept, reduced, or delayed based on user state and task-related conditions. Evaluation metrics such as confusion matrix, classification report, and accuracy score were used to measure model performance.

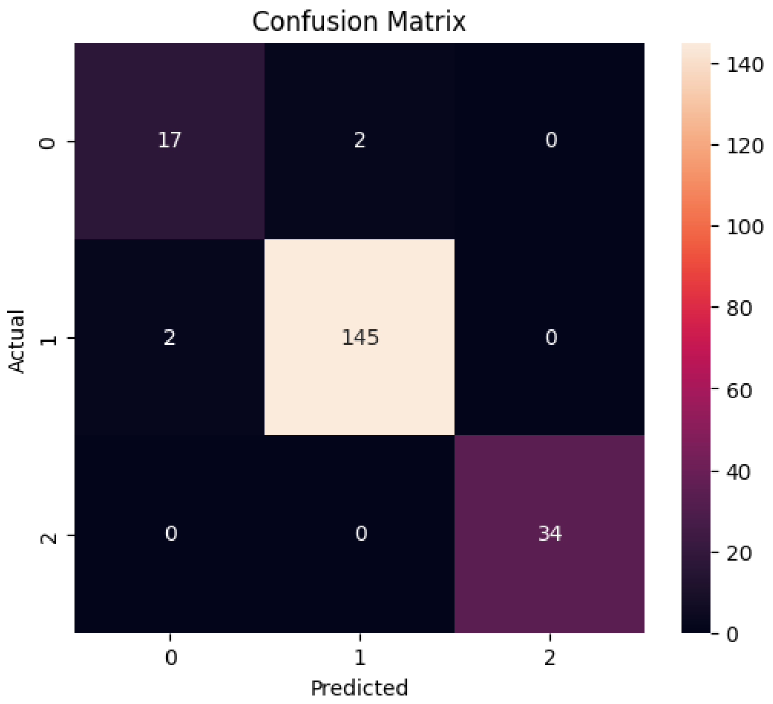

Figure 1: Dataset Correlation Heatmap

The heatmap was used to analyze relationships between dataset features and understand important productivity-related patterns.

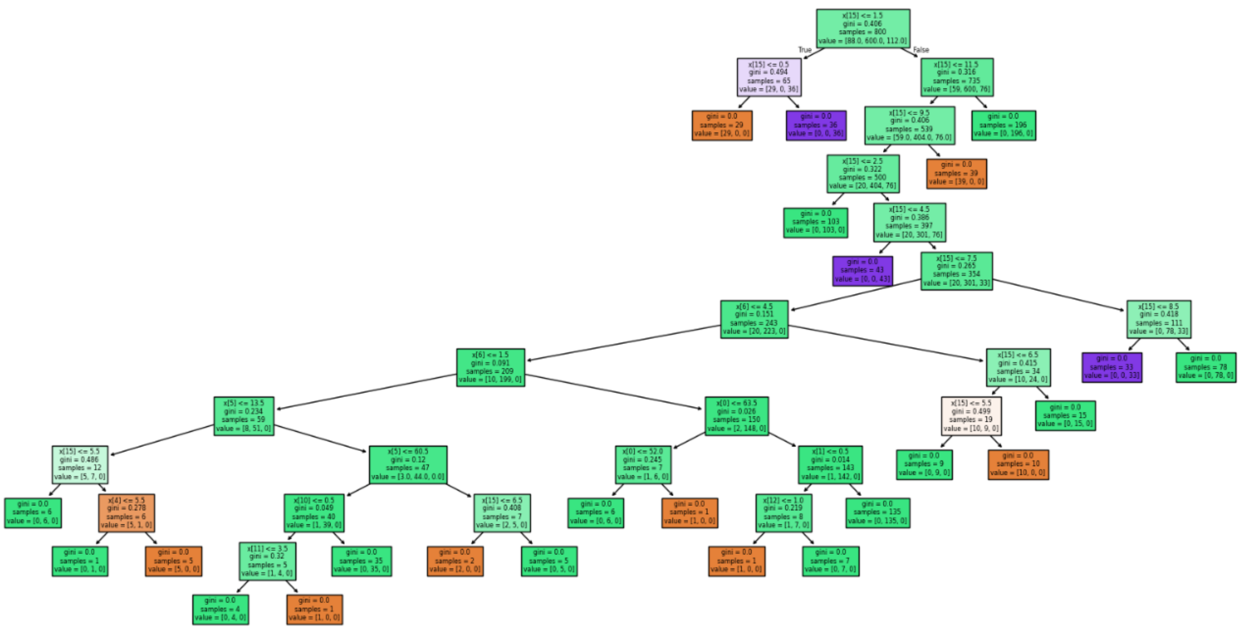

Figure 2: Confusion Matrix

The confusion matrix was used to evaluate prediction accuracy and classification performance of the machine learning model.

The developed prototype provides an interactive interface where users can enter multiple tasks along with emotional and workload-related details. The system calculates total daily workload and intelligently detects overload conditions when the total schedule exceeds 24 hours. Based on user mood, stress level, energy level, sleep hours, task priority, and task difficulty, the system generates personalized productivity balancing suggestions.

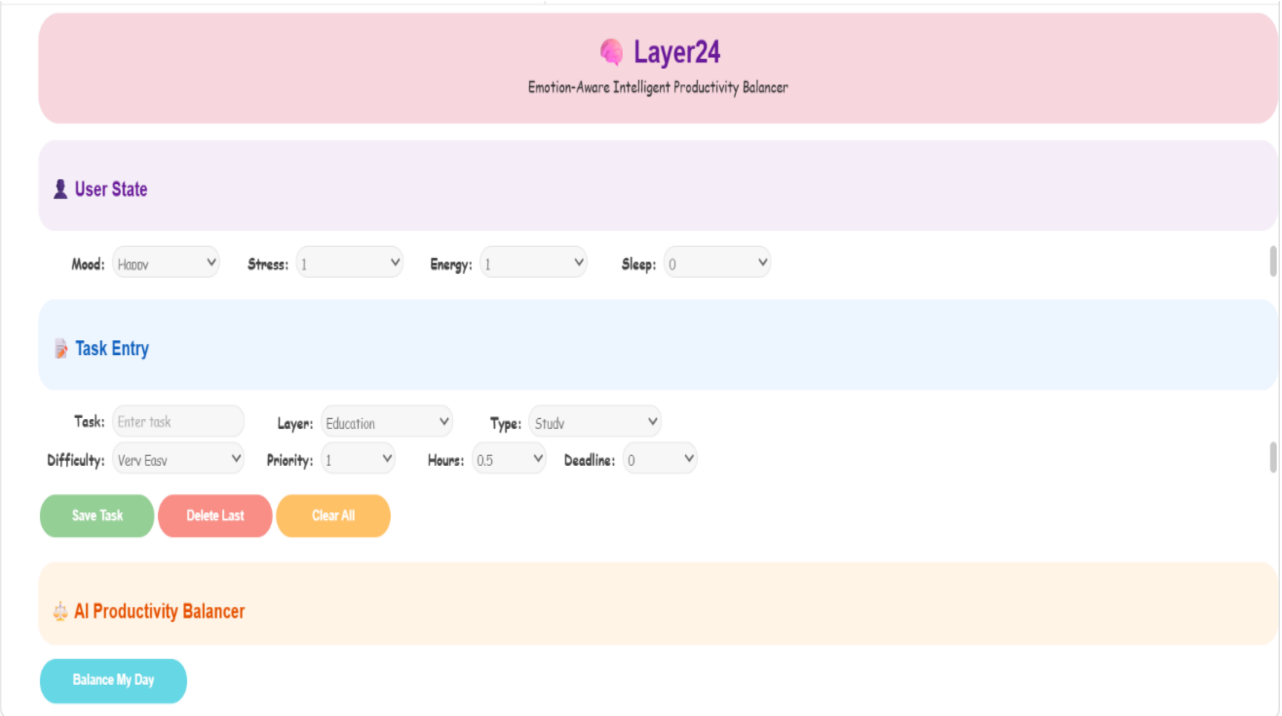

Figure 3: Layer24 Prototype Interface

The developed prototype provides a user-friendly interface for intelligent task balancing and overload detection.

Experimental results showed that the proposed system can support more balanced and human-centered productivity management compared to traditional task management systems. The prototype successfully demonstrated intelligent task balancing and adaptive productivity recommendations.

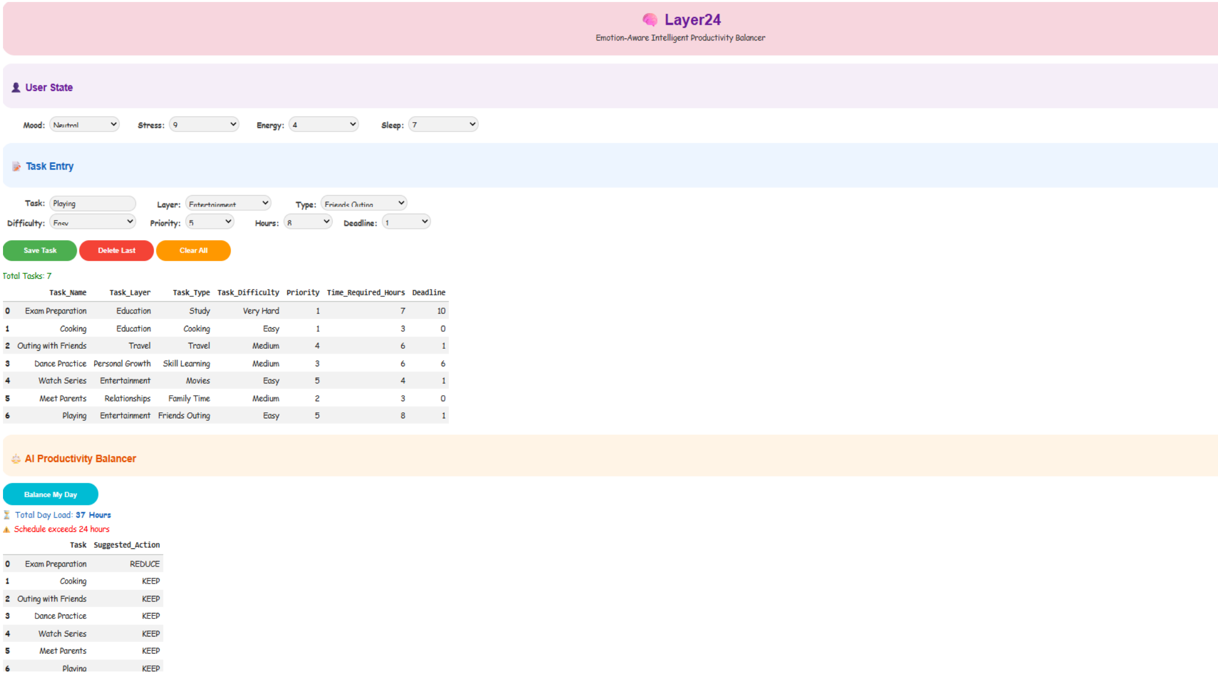

Figure 4: Intelligent Task Balancing Output

The output demonstrates personalized task balancing recommendations generated based on emotional and workload-related conditions.

5. CONCLUSION

This research presented Layer24, an emotion-aware intelligent productivity balancing system developed using machine learning techniques. The system combines user emotional state and task-related information to provide intelligent task management recommendations such as KEEP, REDUCE, and DELAY.

The proposed approach focuses not only on productivity but also on maintaining mental and physical well-being. A custom dataset was created and processed using data preprocessing techniques, and a machine learning model was trained to generate adaptive productivity suggestions. The developed prototype successfully demonstrated intelligent overload detection and human-centered task balancing.

The results of this research show that integrating emotional and contextual factors into productivity systems can improve daily task management and support healthier work-life balance. Future improvements may include advanced AI models, real-time monitoring, and mobile application integration.

6. REFERENCES

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